

Hydrogen, oxygen, and nitrogen

Among the non-metal elements, these three gases are vital to us in different ways. A mixture of nitrogen and oxygen makes up most of the air we breathe, while hydrogen is the most abundant element in the universe.

Each of these gases has atoms that go in pairs: they exist as molecules of two atoms. That is why hydrogen is written as H_2 , oxygen as O_2 , and nitrogen as N_2 . All three elements are found in compounds, such as DNA and proteins, that are vital for all forms of life on Earth.

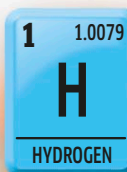
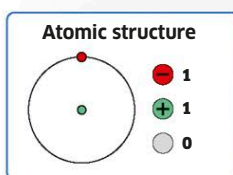
HYDROGEN

Hydrogenium

Discovered: 1766

Hydrogen is the simplest of all the elements.

Its lightest, and most common, isotope has atoms made of a single proton and a single electron, but no neutrons. Hydrogen gets its name from the Greek *hydro* and *genes* meaning “water forming;” when it reacts with oxygen it makes water, or H_2O .



Hydrogen in the universe

Although rare in Earth's atmosphere, hydrogen makes up more than 88 percent of all matter in the universe. Our sun is not much more than a ball of very hot hydrogen. The hydrogen fuses together to produce helium (see p.41), the second element in the periodic table. In the process, a vast amount of energy is produced.

Hydrogen as fuel

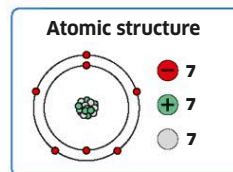
A very reactive element that will burn easily, hydrogen can be used as a fuel. When mixed with oxygen, it forms an explosive mixture. The rocket of a spacecraft uses liquid hydrogen, mixed with liquid oxygen, as fuel. In fuel cells, used in electric cars, the chemical reaction between hydrogen and oxygen is converted to electricity. This combustion reaction produces only water, not water and carbon dioxide as in gasoline-fueled engines, making it an environmentally friendly fuel.

NITROGEN

Nitrogenium

Discovered: 1772

In a nitrogen molecule (N_2), the two atoms are held together with a strong triple bond. The molecule is hard to break apart, which means nitrogen does not react readily with other substances. It is a very common element, making up 78 percent of the air on Earth. It is extremely useful, too. We need it in our bodies and, as part of the nitrogen cycle (see p.186), it helps plants to grow. Where plants and crops need extra help, it is added to fertilizers.



Explosive stuff

Molecules of nitrogen are not reactive, but many compounds containing nitrogen react very easily. These are found in many explosives, such as TNT, dynamite, and gunpowder, and in fireworks, too. On its own, compressed nitrogen gas is used to safely but powerfully blast out paintballs in paintball guns.

Liquid nitrogen

Nitrogen only condenses to liquid if it is cooled to $-321^{\circ}F$ ($-196^{\circ}C$). This means that it is extremely cold in liquid form, instantly freezing anything it comes into contact with. This is useful for storing sensitive blood samples, cells, and tissue for medical use.

